

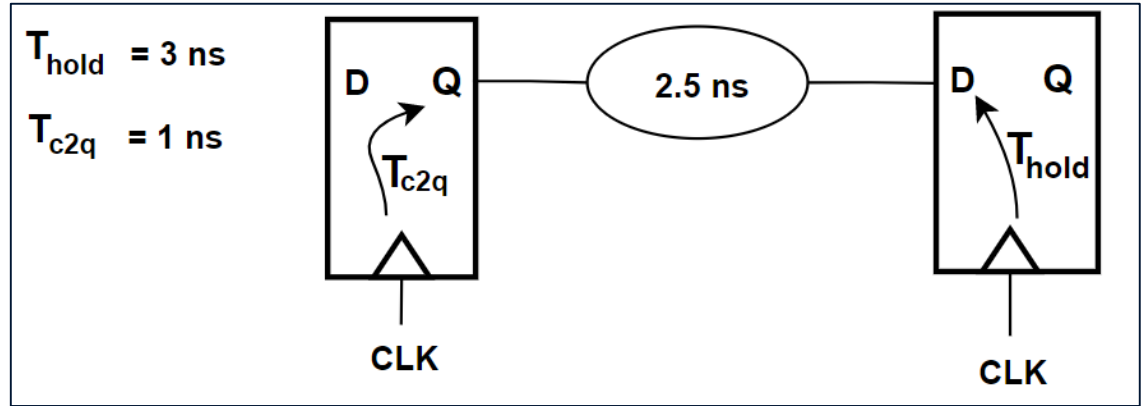
Example 1

- **Assumptions**

1. Interconnect Delays not considered
2. Clock Uncertainty not considered
3. One Operating Condition

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



Example 2

- **Assumptions**

1. Interconnect Delays not considered
2. **Clock Uncertainty is considered**
3. One Operating Condition

- **Required**

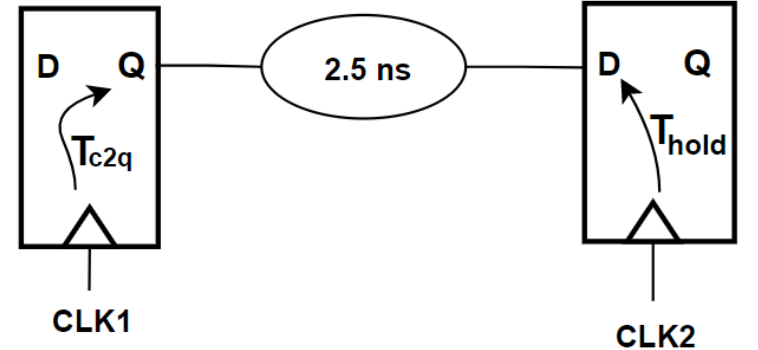
1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack

$$T_{\text{hold}} = 3 \text{ ns}$$

$$T_{\text{c2q}} = 1 \text{ ns}$$

Skew

$$\text{CLK 1_2} = 0.7 \text{ ns}$$



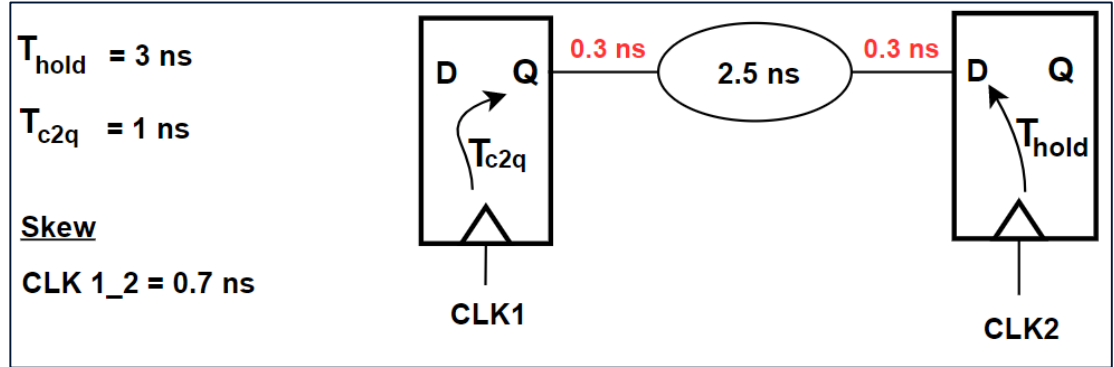
Example 3

- Assumptions**

1. Interconnect Delays is considered
2. Clock Uncertainty is considered
3. One Operating Condition

- Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



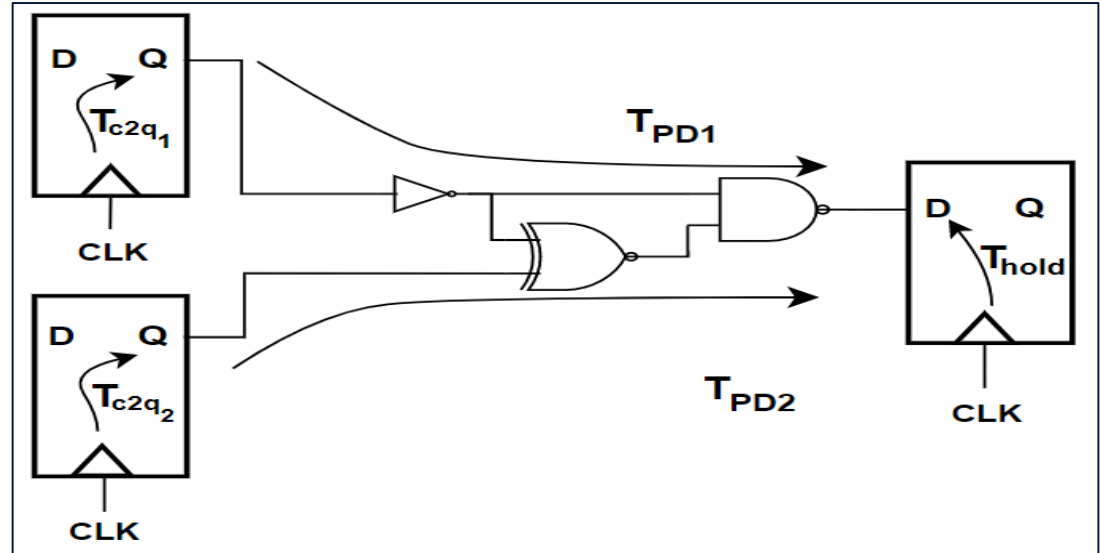
Example 4

- Assumptions**

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

- Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack
3. if not calculate the negative slack and suggest a solution to fix hold violation



Delays	Inverter	XNOR	NAND	Tcq1	Tcq2	Thold
Max	0.5	1.5	1	0.4	0.35	2.2
Min	0.45	1.3	0.95	0.3	0.3	2